

TRDP II

Explaining BERT Sentiment Predictions with Partition SHAP

A token- and word-level analysis of an SST-2 classifier

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This report documents the SHAP experiment implemented as a continuation of the earlier RFEM analysis. The original BERT checkpoint, sentence set, label convention, maximum sequence length, and masking principle were retained so that differences in the explanations arise from the attribution method rather than from a changed classifier.

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Abstract

This study applies Partition SHAP to a BERT classifier fine-tuned for binary sentiment analysis. The experiment was motivated by the failure of an earlier attention-based recursive feature elimination procedure to retain several semantically important words. The classifier output explained here is the difference between the positive and negative logits. A SHAP text masker creates coalitions of available and unavailable tokenizer features; unavailable features are represented by BERT’s [MASK] token, and every perturbed sentence is evaluated by the unchanged classifier. Token-level contributions are subsequently aggregated over BERT WordPieces to obtain readable word-level explanations.

The evaluation used ten qualitative sentences carried over from the previous experiment. The model classified all 10 sentences correctly. For every sentence, the SHAP base value plus the sum of the feature attributions reconstructed the model margin, with a maximum stored additivity residual of $0.00e + 00$. Masking the highest-ranked decision-supporting words produced a mean comprehensiveness drop of 5.3107, while retaining only those words produced a mean sufficiency gap of 0.8927. The explanations recover clear sentiment cues such as *beautiful*, *outstanding*, *boring*, *weak*, and *captivating*. They also expose interaction-dependent assignments that should not be read as context-free lexical sentiment.

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1 Introduction

The previous TRDP experiment investigated whether an attention-derived recursive feature elimination method could identify the words responsible for BERT’s sentiment decisions. That approach did not behave reliably: in several cases, words carrying the clearest sentiment were assigned low importance or were removed before less informative tokens. The present experiment therefore changes the explanation method while leaving the predictive model and qualitative inputs unchanged.

SHAP was selected because it defines feature attribution through changes in a model output across coalitions of available features [1]. This is substantively different from treating attention weights as feature importance. For text, the relevant question is counterfactual: how does the classifier’s score change when a token or group of tokens is unavailable while other tokens remain visible? Partition SHAP answers this question efficiently by respecting a hierarchical clustering of features and estimating Owen-style values rather than enumerating all 2^M coalitions.

The objectives of the experiment are:

- (1) to obtain signed local attributions for each BERT tokenizer feature;
- (2) to aggregate WordPiece attributions into readable word-level values;
- (3) to verify the additive reconstruction of the explained classifier score;
- (4) to test explanation faithfulness through comprehensiveness and sufficiency;
- (5) to compare the resulting explanations with the qualitative behavior observed in the earlier RFEM study.

2 Model and explanation methodology

2.1 BERT sentiment classifier

Let the input sentence be x . BERT’s tokenizer maps it to a sequence

$$\tau(x) = (t_0, t_1, \dots, t_{M+1}), \quad (1)$$

where $t_0 = [\text{CLS}]$, $t_{M+1} = [\text{SEP}]$, and the intervening features may be complete words, punctuation, or WordPiece fragments. The fine-tuned classifier returns two logits,

$$\mathbf{z}_\theta(x) = (z_{\text{NEG}}(x), z_{\text{POS}}(x)). \quad (2)$$

Class probabilities are obtained with the softmax function:

$$p_\theta(y = c \mid x) = \frac{\exp(z_c(x))}{\exp(z_{\text{NEG}}(x)) + \exp(z_{\text{POS}}(x))}. \quad (3)$$

The predicted class is $\hat{y} = \arg \max_c p_\theta(y = c \mid x)$. The checkpoint is `textattack/bert-base-uncased-SST-2`, which is based on the BERT architecture [2] and fine-tuned for the SST-2 task derived from the Stanford Sentiment Treebank [3]. The implementation keeps the previous TRDP label convention: index 0 is negative and index 1 is positive.

2.2 Scalar target explained by SHAP

SHAP requires the explained model to return a scalar for each perturbed input. The wrapper therefore uses the signed logit margin

$$f(x) = z_{\text{POS}}(x) - z_{\text{NEG}}(x). \quad (4)$$

A positive margin favors POS and a negative margin favors NEG. This choice also fixes the sign convention of every attribution: $\phi_i > 0$ pushes the decision toward positive sentiment, and $\phi_i < 0$ pushes it toward negative sentiment. Using the logit margin avoids probability saturation and preserves the classifier’s unbounded evidence scale.

2.3 Coalitions and the SHAP text masker

For M explainable tokenizer features, a coalition is represented by a binary vector $\mathbf{m} \in \{0, 1\}^M$. A value of one marks a feature as available; a value of zero marks it as unavailable. The reconstruction function $h_x(\mathbf{m})$ maps the coalition to a perturbed sentence:

$$h_x(\mathbf{m})_i = \begin{cases} t_i, & m_i = 1, \\ [\text{MASK}], & m_i = 0. \end{cases} \quad (5)$$

The corresponding coalition value is

$$v_x(S) = f(h_x(\mathbf{1}_S)), \quad S \subseteq \{1, \dots, M\}. \quad (6)$$

Consecutive mask tokens were not collapsed. This preserves tokenizer positions and keeps the perturbation convention aligned with the earlier BERT experiment. The [CLS] and [SEP] tokens remain part of model processing, whereas punctuation and special tokens are excluded from content-word rankings.

2.4 Shapley and Partition SHAP attribution

For unrestricted coalitions, the Shapley value of feature i is the weighted average of its marginal contribution over all subsets not containing i :

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} [v_x(S \cup \{i\}) - v_x(S)], \quad (7)$$

where $N = \{1, \dots, M\}$. Exact evaluation is exponential in M . The Partition explainer instead builds a hierarchical feature tree $\mathcal{T}$ and evaluates coalitions consistent with that hierarchy. In permutation form, the resulting Owen-style attribution may be written as

$$\phi_i^{\mathcal{T}} = \frac{1}{|\Pi_{\mathcal{T}}|} \sum_{\pi \in \Pi_{\mathcal{T}}} [v_x(P_i^{\pi} \cup \{i\}) - v_x(P_i^{\pi})], \quad (8)$$

where $\Pi_{\mathcal{T}}$ is the set of hierarchy-consistent permutations and P_i^{π} denotes the features preceding i in permutation π . The implementation allowed at most 500 model evaluations per sentence and processed perturbations in batches of 32.

The local explanation is the additive model

$$g(\mathbf{m}) = \phi_0 + \sum_{i=1}^M \phi_i m_i, \quad \phi_0 = v_x(\emptyset). \quad (9)$$

For the original sentence, local accuracy requires

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i. \quad (10)$$

The recorded additivity residual is

$$\epsilon_{\text{add}} = f(x) - \left(\phi_0 + \sum_{i=1}^M \phi_i \right). \quad (11)$$

2.5 WordPiece aggregation

BERT may split a surface word into several WordPieces. If I_w is the set of tokenizer positions belonging to word w , its word-level attribution is the signed sum

$$\Phi_w = \sum_{i \in I_w} \phi_i. \quad (12)$$

For example, the fragments **res**, **##ili**, and **##ence** are displayed as *resilience*. Summation preserves the total contribution assigned to the original tokenizer features.

2.6 Faithfulness measures

Faithfulness was evaluated in the direction of the model’s predicted class. Define

$$d = \begin{cases} +1, & \hat{y} = \text{POS}, \\ -1, & \hat{y} = \text{NEG}, \end{cases} \quad q(x) = d f(x). \quad (13)$$

Thus $q(x) > 0$ indicates support for the current prediction in both classes. Words are ranked by $d\Phi_w$. For a fraction α , the number of selected words is

$$k_\alpha = \max(1, \lceil \alpha W \rceil), \quad (14)$$

where W is the number of content-word groups. Let R_α be the top k_α groups. Comprehensiveness measures how much the decision score falls after those groups are masked:

$$C_\alpha = q(x) - q(x \setminus R_\alpha). \quad (15)$$

Higher C_α indicates that the selected words were important for the original prediction. The sufficiency gap measures the loss when only those words remain visible:

$$S_\alpha = q(x) - q(x_{R_\alpha}). \quad (16)$$

Lower S_α is preferred. Negative values can occur when the retained subset supports the predicted class more strongly than the complete sentence. These definitions follow the general rationale used in rationale evaluation [4] while retaining the same text masker used by the explainer.

2.7 Cross-sentence comparison

The global plot is descriptive rather than a dataset-level global explanation. For each word type w observed n_w times in the ten sentences, it reports

$$G(w) = \frac{1}{n_w} \sum_{j=1}^{n_w} |\Phi_{w,j}|. \quad (17)$$

Absolute values are used because strong evidence for either class is important.

3 Implementation workflow

Figure 1 summarizes the implemented path from the raw sentence to word-level values. SHAP supplies coalitions to the text masker; the model wrapper evaluates every resulting text under `torch.inference_mode()`, and Partition SHAP combines the score differences. WordPiece aggregation takes place only after token-level SHAP values have been computed.

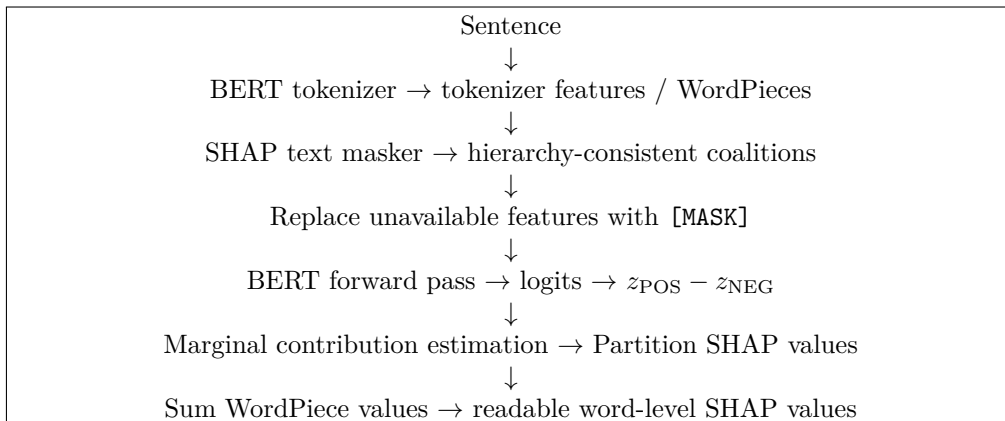


Figure 1: Processing sequence followed by the implementation.

The implementation is divided into five source files. The model wrapper isolates tokenization and inference; token aggregation handles feature alignment and WordPiece joining; faithfulness evaluation applies ranked masks; plotting code contains all visual output; and the run script coordinates configuration, I/O, and execution. This separation made it possible to test model parity and SHAP arithmetic independently of plotting and report generation.

4 Experimental setup

4.1 Data

The qualitative set contains ten manually specified sentences, denoted S1–S10. Six have positive gold labels and four have negative gold labels. Several were constructed to contain contrast or mixed evidence, including S3 (*slow* versus *outstanding*) and S8 (*weak* versus *captivating*). The complete set is reproduced in Appendix A.

4.2 Configuration and reproducibility

The run used Python 3.12.12, PyTorch 2.12.0, Transformers 5.9.0, SHAP 0.52.0, NumPy 2.5.3, and Matplotlib 3.10.9 on CPU. Maximum sequence length was 128, `max_evals` was 500 per sentence, and the model evaluation batch size was 32. Random seeds for NumPy and PyTorch were set to 42. The complete run took 39.70 seconds, evaluated 3,814 perturbed texts in 253 forward batches, and stored the configuration alongside the numerical results.

5 Results

5.1 Classification and additivity

Table 1 reports the model decision and SHAP decomposition for every sentence. All 10 predictions agree with the gold labels. The predicted-class probabilities exceed 0.999 for most examples; S2, S3, S4, S7, S8, and S9 remain above 0.999 or close to it despite mixed wording in some cases. These ten examples are qualitative probes, so the result should not be interpreted as an estimate of held-out accuracy.

Table 1: Sentence-level classification and SHAP additivity results.

ID	Gold	Pred.	Correct	$p(\hat{y} \mid x)$	$f(x)$	ϕ_0	$\sum_i \phi_i$	ϵ_{add}
S1	POS	POS	1	0.999738	8.2464	3.5955	4.6509	0.0e+00
S2	NEG	NEG	1	0.999214	-7.1473	2.6621	-9.8094	0.0e+00
S3	POS	POS	1	0.999055	6.9631	3.5955	3.3675	0.0e+00
S4	NEG	NEG	1	0.999311	-7.2796	2.6621	-9.9416	0.0e+00
S5	POS	POS	1	0.999716	8.1665	3.5995	4.5670	0.0e+00
S6	POS	POS	1	0.999731	8.2199	2.6621	5.5579	0.0e+00
S7	NEG	NEG	1	0.999366	-7.3636	3.0872	-10.4508	0.0e+00
S8	POS	POS	1	0.999030	6.9375	3.5955	3.3420	0.0e+00
S9	NEG	NEG	1	0.999217	-7.1515	3.5955	-10.7470	0.0e+00
S10	POS	POS	1	0.999771	8.3806	3.5995	4.7811	0.0e+00

The maximum absolute residual in Equation (11) is $0.00e + 00$ at the stored precision. This verifies that the saved attributions correspond to the same scalar margin returned by the BERT wrapper. It does not, by itself, establish that the masking baseline is linguistically ideal or that every individual value has a context-free semantic interpretation.

5.2 Dominant local attributions

Table 2 identifies the largest positive and negative word-level attribution in each sentence. The signs refer to the POS-minus-NEG margin, not to correctness. Thus a strongly negative value is supporting evidence when the model predicts NEG.

Table 2: Largest positive and negative word-level attributions in each sentence.

ID	Gold	Prediction	Largest positive attribution	Largest negative attribution
S1	POS	POS	beautiful (+2.2569)	of (-0.1773)
S2	NEG	NEG	and (+2.3351)	waste (-5.6937)
S3	POS	POS	outstanding (+3.7479)	slow (-1.0241)
S4	NEG	NEG	a (+0.3156)	dull (-4.0125)
S5	POS	POS	brilliant (+4.0473)	the (-0.5056)
S6	POS	POS	breathtaking (+1.8503)	emotion (-1.7249)
S7	NEG	NEG	charm (+1.5548)	boring (-3.5674)
S8	POS	POS	captivating (+4.7673)	weak (-3.8786)
S9	NEG	NEG	a (+0.5032)	hollow (-3.6939)
S10	POS	POS	entertaining (+2.8358)	to (-0.4039)

The explanations capture the main contrastive cues. In S1, *beautiful* has a SHAP value of +2.2569, followed by *portrait* at +1.0118 and *resilience* at +0.8187. In S3, *outstanding* contributes +3.7479, while *slow* contributes -1.0241. In S7, *boring* contributes -3.5674 and *devoid* contributes -1.6886. The strongest contrast occurs in S8: *captivating* contributes +4.7673, *delivers* contributes +2.1066, and *weak* contributes -3.8786. These results address the central problem observed with RFEM: the explanation now assigns substantial magnitude to sentiment-bearing content words rather than eliminating them as unimportant.

Not every attribution is lexically intuitive. In S7, *or* receives a large negative contribution, while *charm* receives a positive contribution even though the complete sentence is strongly negative. In S6, *emotion* is negative despite the positive prediction. Such values reflect conditional model behavior under the chosen coalitions and masking baseline. They should be read as local interaction-dependent allocations, not as a sentiment dictionary.

5.3 Cross-sentence importance

Figure 2 ranks the twenty word types with the highest mean absolute word-level SHAP magnitude. Because the experiment contains only ten sentences, the plot summarizes this selected set and is not a population-level estimate.

The largest magnitudes belong primarily to overt evaluative words, including *waste*, *captivating*, *brilliant*, *outstanding*, *hollow*, *boring*, and *dull*. This ordering is consistent with the local plots, although averaging absolute values removes both sentence context and attribution direction.

5.4 Faithfulness

Table 3 gives the mean faithfulness scores for each selected fraction. Comprehensiveness increases as more highly ranked words are removed, rising from approximately 2.38 at $\alpha = 0.1$ to 9.22 at $\alpha = 0.5$. The sufficiency gap decreases from approximately 1.34 to 0.40 as more words are retained. Both trends are consistent with the expected behavior of a useful ranking.

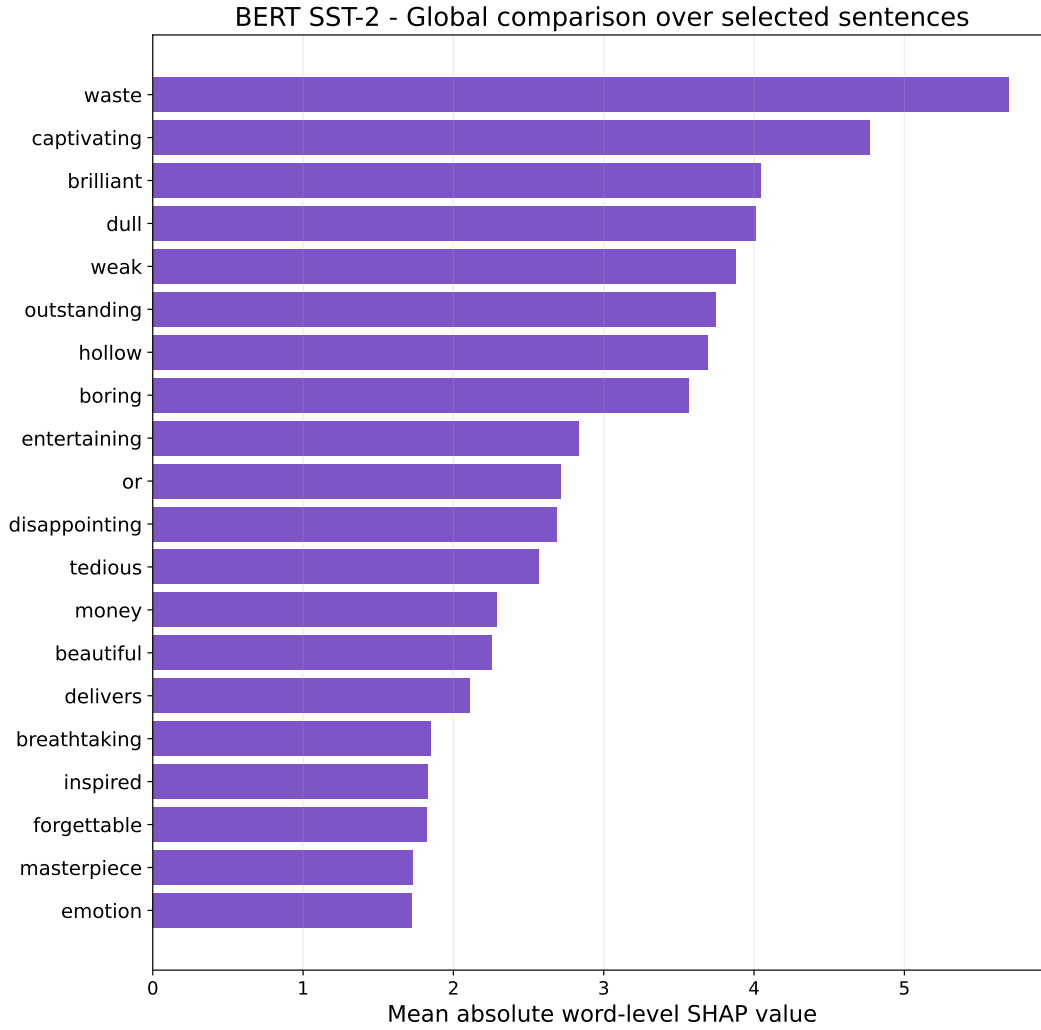


Figure 2: Mean absolute word-level SHAP value over the ten qualitative sentences.

Table 3: Faithfulness results averaged over the ten sentences. Standard deviations are population standard deviations.

Fraction α	Mean selected words	Mean comp.	SD comp.	Mean suff. gap	SD suff. gap
0.1	1.50	2.3791	2.8969	1.3380	0.8157
0.2	2.50	3.6865	3.8714	1.0390	0.4936
0.3	3.50	5.9559	3.1109	0.7926	0.6147
0.5	5.40	9.2213	2.6117	0.4013	0.6097

Across all forty fraction-sentence combinations, the mean comprehensiveness drop is 5.3107 and the mean sufficiency gap is 0.8927. The sentence-wise means in Table 4 show that the strength of this behavior varies across examples.

Table 4: Mean faithfulness scores by sentence across $\alpha \in \{0.1, 0.2, 0.3, 0.5\}$.

ID	Mean comprehensiveness drop	Mean sufficiency gap
S1	3.2323	1.2103
S2	7.3672	0.8264
S3	9.1041	-0.1540
S4	5.4999	0.6360
S5	4.8566	0.9113
S6	2.4668	1.4390
S7	1.4940	1.5450
S8	10.6432	0.3396
S9	4.3473	0.8867
S10	4.0954	1.2871

The detailed masks and scores are given in Appendix D. A negative sufficiency gap in a few rows is not an error: it means that retaining the selected subset increased the decision-direction score relative to the full sentence, usually because conflicting evidence was removed.

6 Discussion

The SHAP experiment offers three improvements over the failed RFEM analysis. First, it explains changes in an explicitly defined model output rather than assuming that attention magnitude is equivalent to causal importance. Second, it evaluates a feature in multiple coalition contexts, allowing interactions to affect the final allocation. Third, it provides a direct additivity check and a separate masking-based faithfulness evaluation.

The results nevertheless depend on methodological choices. The most important choice is the missing-feature representation. BERT was fine-tuned on ordinary SST-2 sentences, whereas the masker creates inputs containing one or more [MASK] tokens. The classifier’s behavior on those inputs may not match its behavior on naturally occurring omissions. Partition SHAP also restricts coalitions through a hierarchy; its values are Owen-style allocations and need not match unrestricted exact Shapley values. Increasing the evaluation budget reduces approximation variability but does not remove baseline dependence.

The sample is intentionally small and qualitative. It is suitable for inspecting whether explanations recover known cues and handle contrastive examples, but it cannot support claims about general explanation accuracy across SST-2. A larger evaluation should include the SST-2 development set, perturbation baselines other than [MASK], repeated runs at several budgets, and comparisons with alternative attribution methods such as Integrated Gradients.

7 Conclusion

Partition SHAP was successfully integrated with the existing BERT SST-2 classifier without changing the model, label mapping, or qualitative sentence set. The method generated signed tokenizer-level explanations, preserved local additivity for the POS-minus-NEG logit margin, and produced readable word-level values through WordPiece aggregation. All ten predictions

were correct, and the main sentiment-bearing words generally received the largest attribution magnitudes. Faithfulness scores changed in the expected direction as larger fractions of the ranked words were removed or retained.

The experiment therefore provides a more defensible feature-attribution analysis than the earlier attention-based elimination procedure. Its conclusions remain local to the selected masker, model, and ten-sentence sample; the values should be treated as model-behavior explanations under controlled perturbations rather than as intrinsic linguistic properties of individual words.

A Qualitative sentence set

Table 5: Complete qualitative sentence set used in the experiment.

ID	Gold	Sentence
S1	POS	The film is a beautiful and moving portrait of human resilience.
S2	NEG	This movie is an absolute waste of time and money.
S3	POS	It’s a bit slow at times but the performances are outstanding.
S4	NEG	A dull, tedious and completely forgettable experience.
S5	POS	The direction is inspired and the acting is nothing short of brilliant.
S6	POS	A masterpiece of storytelling with breathtaking visuals and emotion.
S7	NEG	Painfully boring and utterly devoid of any originality or charm.
S8	POS	The screenplay is weak but the lead actor delivers a captivating turn.
S9	NEG	A hollow and disappointing sequel that betrays everything the original stood for.
S10	POS	Funny, heartfelt and endlessly entertaining from beginning to end.

B Complete sentence-level plots

The following figures reproduce every word-level plot produced by the verified `modular_s1_s10` run. Red bars increase the POS-minus-NEG margin; blue bars decrease it. The titles report the prediction, margin, base value, and additivity residual.

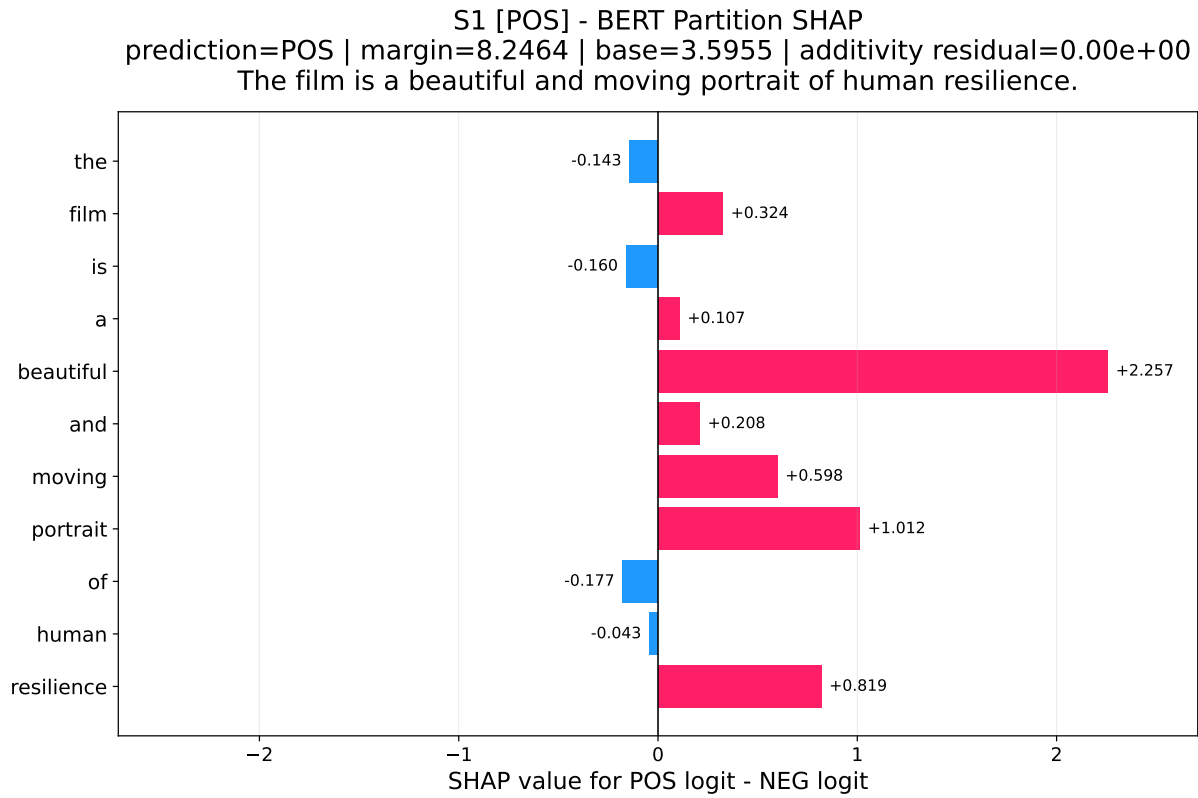


Figure 3: Word-level Partition SHAP explanation for S1.

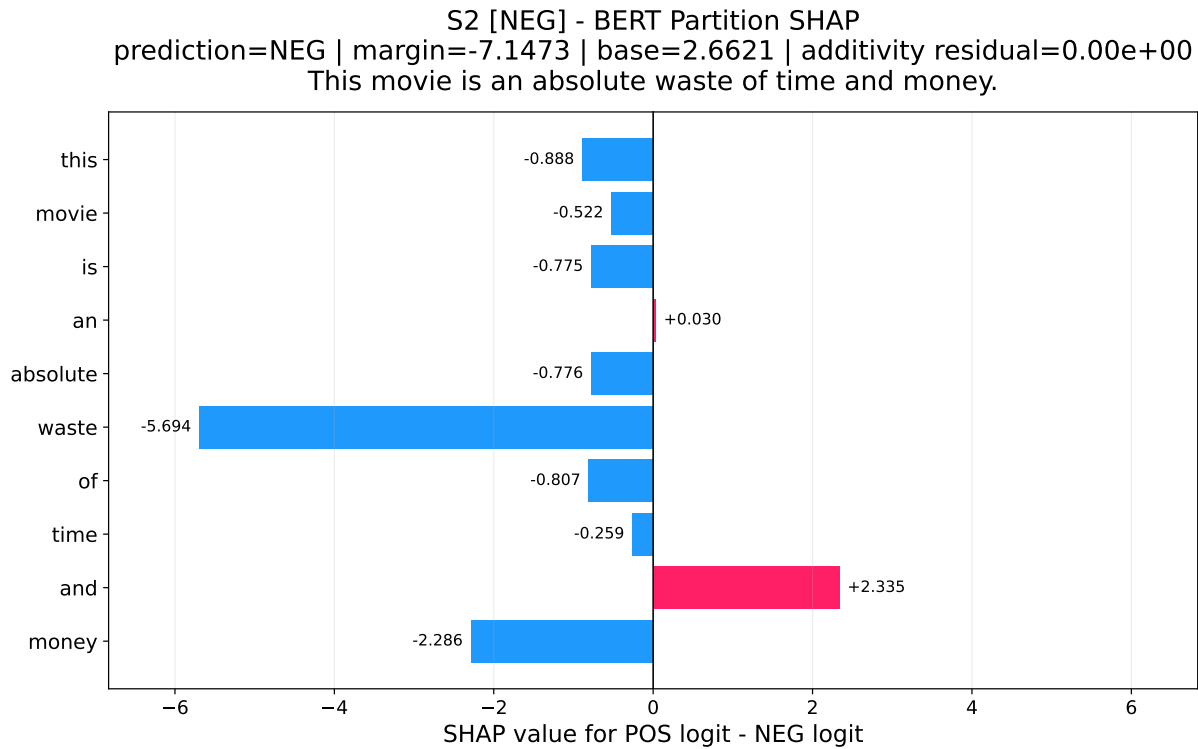


Figure 4: Word-level Partition SHAP explanation for S2.

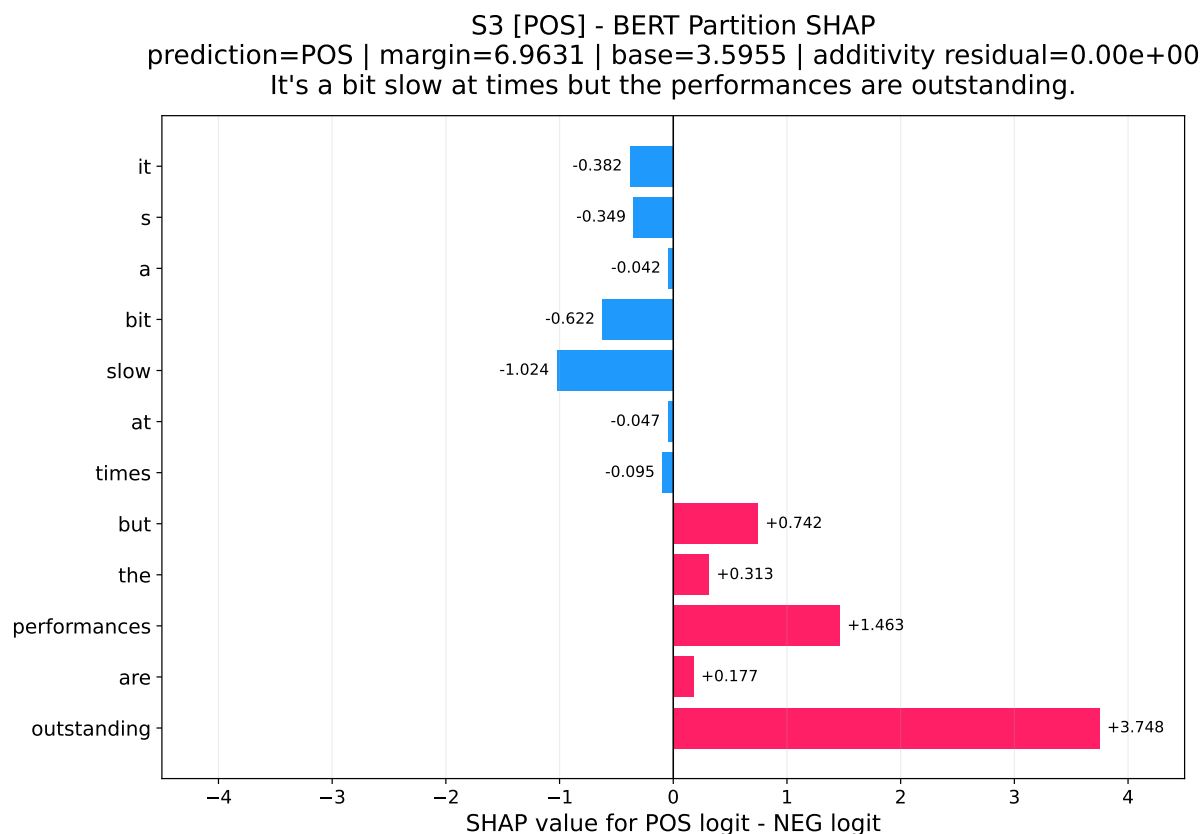


Figure 5: Word-level Partition SHAP explanation for S3.

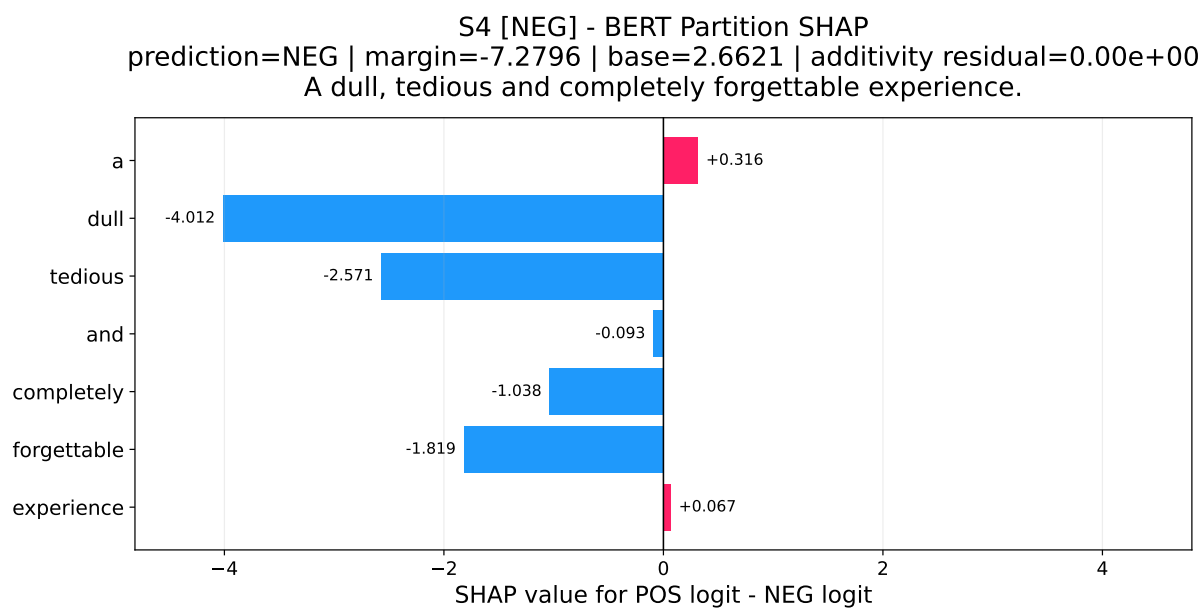


Figure 6: Word-level Partition SHAP explanation for S4.

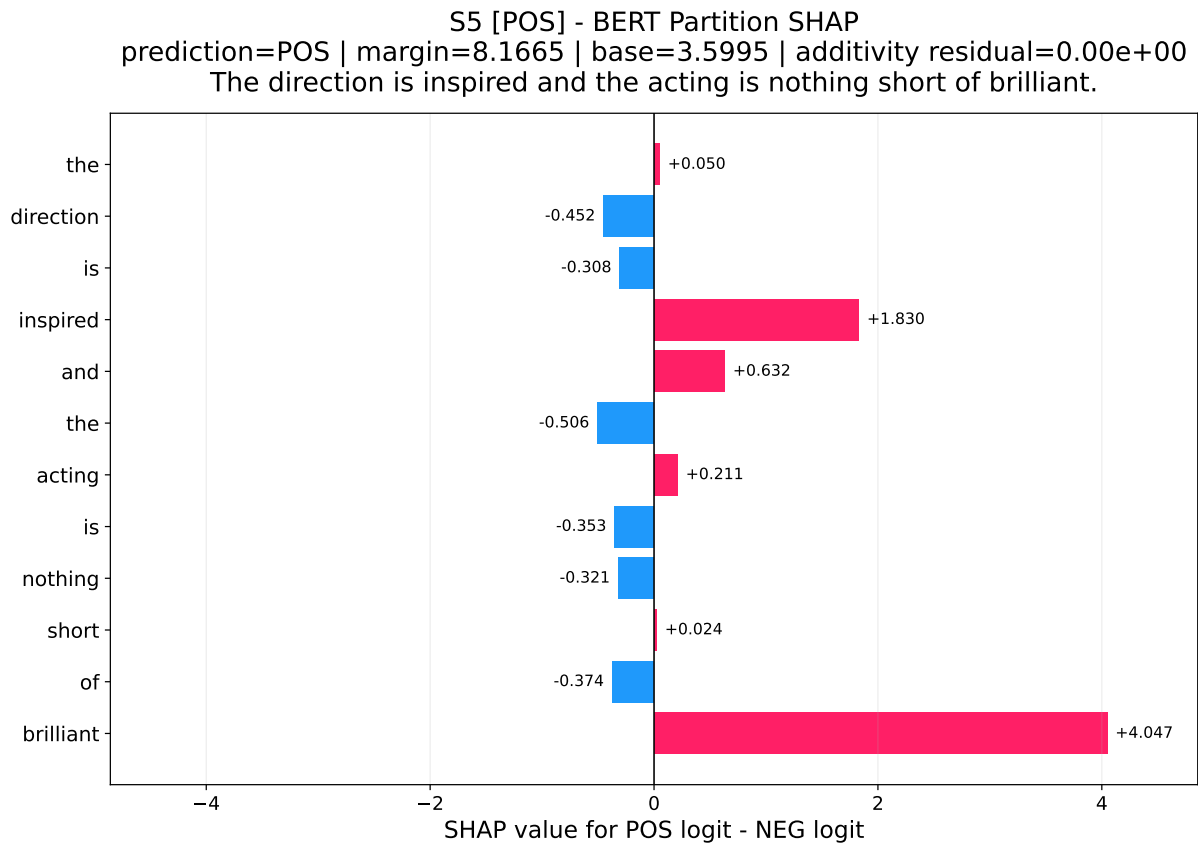


Figure 7: Word-level Partition SHAP explanation for S5.

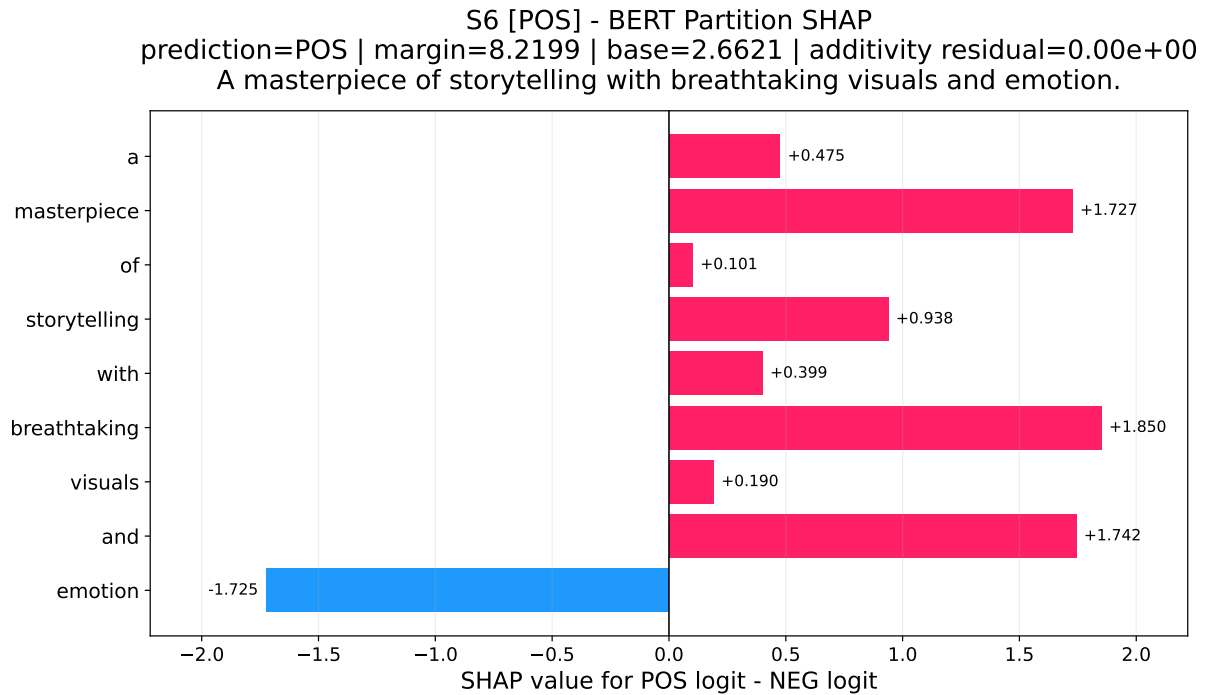


Figure 8: Word-level Partition SHAP explanation for S6.

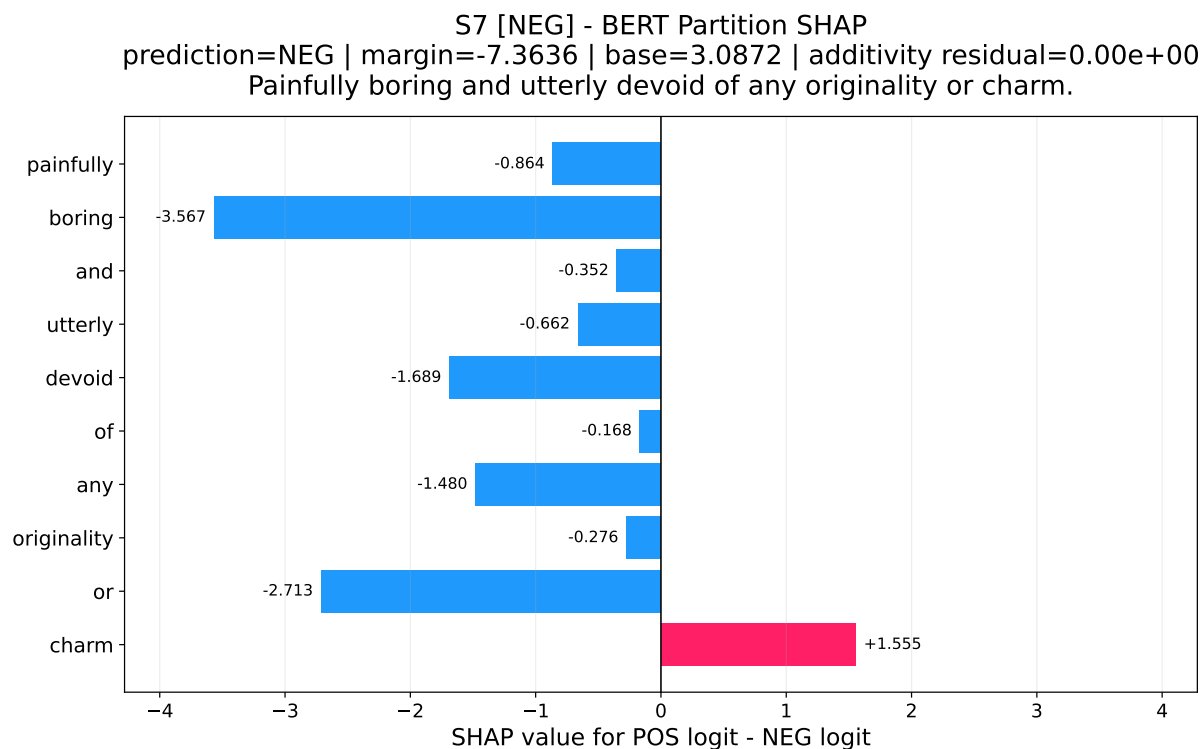


Figure 9: Word-level Partition SHAP explanation for S7.

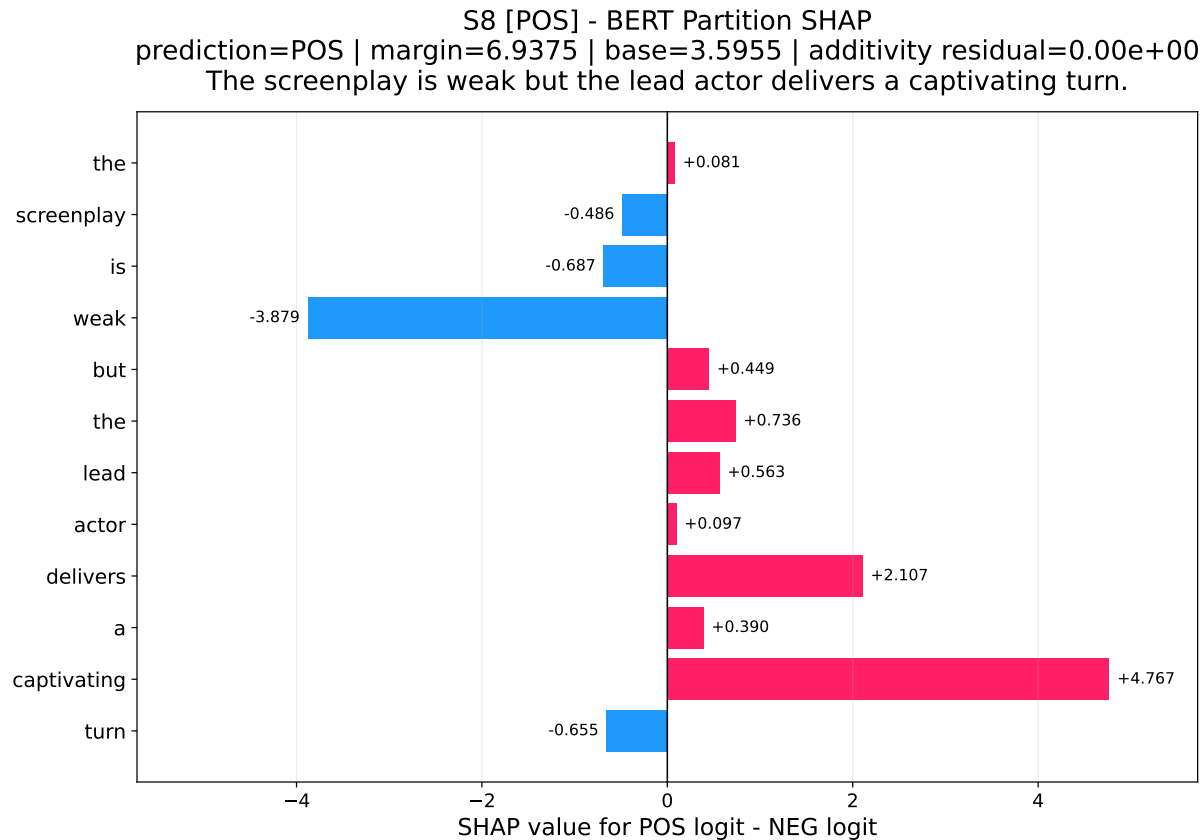


Figure 10: Word-level Partition SHAP explanation for S8.

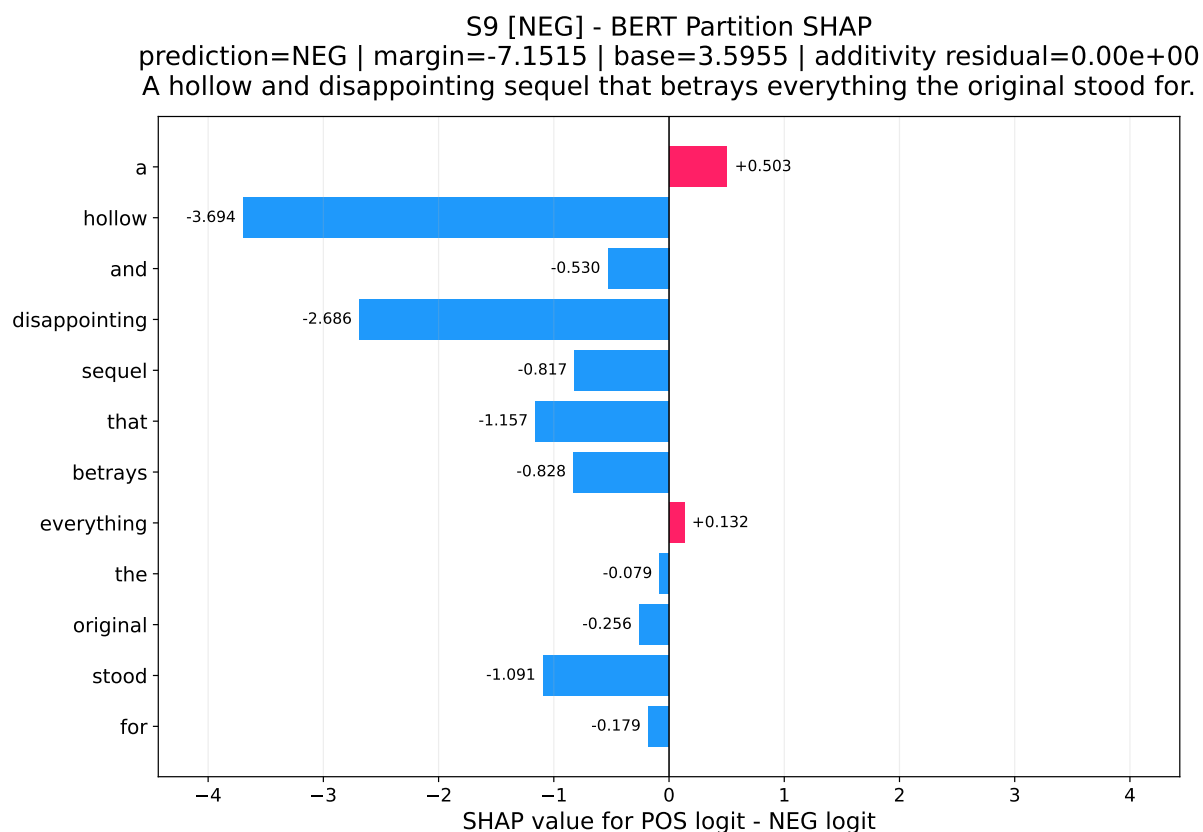


Figure 11: Word-level Partition SHAP explanation for S9.

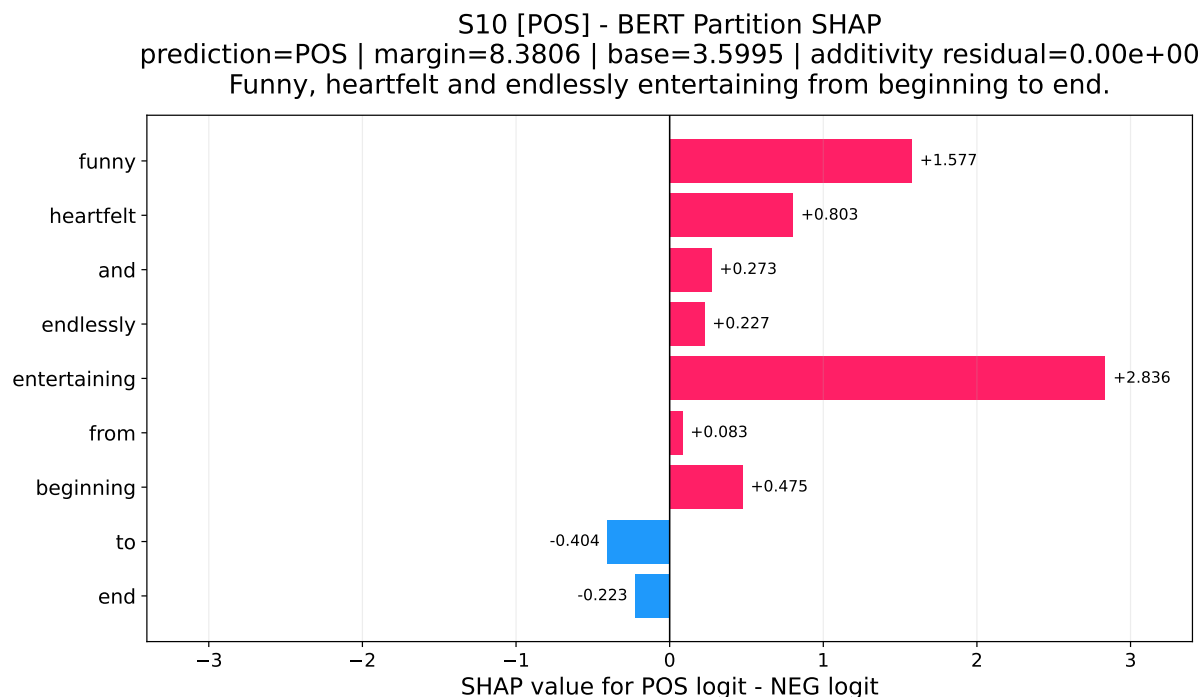


Figure 12: Word-level Partition SHAP explanation for S10.

C Complete word-level numerical results

Table 6: Complete word-level SHAP results. Support is signed in the direction of the predicted class.

ID	Index	Word	Φ_w	$ \Phi_w $	Prediction support
S1	0	the	-0.142536	0.142536	-0.142536
S1	1	film	0.324253	0.324253	0.324253
S1	2	is	-0.160388	0.160388	-0.160388
S1	3	a	0.106850	0.106850	0.106850
S1	4	beautiful	2.256860	2.256860	2.256860
S1	5	and	0.208325	0.208325	0.208325
S1	6	moving	0.598110	0.598110	0.598110
S1	7	portrait	1.011846	1.011846	1.011846
S1	8	of	-0.177266	0.177266	-0.177266
S1	9	human	-0.043383	0.043383	-0.043383
S1	10	resilience	0.818661	0.818661	0.818661
S2	0	this	-0.888101	0.888101	0.888101
S2	1	movie	-0.521932	0.521932	0.521932
S2	2	is	-0.775025	0.775025	0.775025
S2	3	an	0.030125	0.030125	-0.030125
S2	4	absolute	-0.775911	0.775911	0.775911
S2	5	waste	-5.693739	5.693739	5.693739
S2	6	of	-0.807296	0.807296	0.807296
S2	7	time	-0.259404	0.259404	0.259404
S2	8	and	2.335107	2.335107	-2.335107
S2	9	money	-2.285910	2.285910	2.285910
S3	0	it	-0.381619	0.381619	-0.381619
S3	1	s	-0.348592	0.348592	-0.348592
S3	2	a	-0.042201	0.042201	-0.042201
S3	3	bit	-0.621683	0.621683	-0.621683
S3	4	slow	-1.024147	1.024147	-1.024147
S3	5	at	-0.046729	0.046729	-0.046729
S3	6	times	-0.094758	0.094758	-0.094758
S3	7	but	0.741936	0.741936	0.741936
S3	8	the	0.312652	0.312652	0.312652
S3	9	performances	1.462643	1.462643	1.462643
S3	10	are	0.177315	0.177315	0.177315
S3	11	outstanding	3.747872	3.747872	3.747872
S4	0	a	0.315643	0.315643	-0.315643
S4	1	dull	-4.012466	4.012466	4.012466
S4	2	tedious	-2.570508	2.570508	2.570508
S4	3	and	-0.092597	0.092597	0.092597
S4	4	completely	-1.037591	1.037591	1.037591
S4	5	forgettable	-1.819339	1.819339	1.819339
S4	6	experience	0.067303	0.067303	-0.067303
S5	0	the	0.050161	0.050161	0.050161
S5	1	direction	-0.451905	0.451905	-0.451905
S5	2	is	-0.307505	0.307505	-0.307505
S5	3	inspired	1.829674	1.829674	1.829674
S5	4	and	0.631725	0.631725	0.631725
S5	5	the	-0.505583	0.505583	-0.505583
S5	6	acting	0.210538	0.210538	0.210538
S5	7	is	-0.352823	0.352823	-0.352823
S5	8	nothing	-0.320572	0.320572	-0.320572
S5	9	short	0.023784	0.023784	0.023784
S5	10	of	-0.374335	0.374335	-0.374335

Continued on next page

Table 6 – continued

ID	Index	Word	Φ_w	$ \Phi_w $	Prediction support
S5	11	brilliant	4.047340	4.047340	4.047340
S6	0	a	0.474689	0.474689	0.474689
S6	1	masterpiece	1.726510	1.726510	1.726510
S6	2	of	0.100716	0.100716	0.100716
S6	3	storytelling	0.938018	0.938018	0.938018
S6	4	with	0.399469	0.399469	0.399469
S6	5	breathhtaking	1.850310	1.850310	1.850310
S6	6	visuals	0.190360	0.190360	0.190360
S6	7	and	1.742435	1.742435	1.742435
S6	8	emotion	-1.724871	1.724871	-1.724871
S7	0	painfully	-0.863948	0.863948	0.863948
S7	1	boring	-3.567362	3.567362	3.567362
S7	2	and	-0.352303	0.352303	0.352303
S7	3	utterly	-0.662267	0.662267	0.662267
S7	4	devoid	-1.688613	1.688613	1.688613
S7	5	of	-0.168164	0.168164	0.168164
S7	6	any	-1.480106	1.480106	1.480106
S7	7	originality	-0.275503	0.275503	0.275503
S7	8	or	-2.712506	2.712506	2.712506
S7	9	charm	1.554754	1.554754	-1.554754
S8	0	the	0.080639	0.080639	0.080639
S8	1	screenplay	-0.485905	0.485905	-0.485905
S8	2	is	-0.686718	0.686718	-0.686718
S8	3	weak	-3.878561	3.878561	-3.878561
S8	4	but	0.449248	0.449248	0.449248
S8	5	the	0.735832	0.735832	0.735832
S8	6	lead	0.562615	0.562615	0.562615
S8	7	actor	0.097387	0.097387	0.097387
S8	8	delivers	2.106649	2.106649	2.106649
S8	9	a	0.390013	0.390013	0.390013
S8	10	captivating	4.767265	4.767265	4.767265
S8	11	turn	-0.654583	0.654583	-0.654583
S9	0	a	0.503179	0.503179	-0.503179
S9	1	hollow	-3.693891	3.693891	3.693891
S9	2	and	-0.529555	0.529555	0.529555
S9	3	disappointing	-2.685714	2.685714	2.685714
S9	4	sequel	-0.816540	0.816540	0.816540
S9	5	that	-1.157125	1.157125	1.157125
S9	6	betrays	-0.827926	0.827926	0.827926
S9	7	everything	0.131713	0.131713	-0.131713
S9	8	the	-0.079131	0.079131	0.079131
S9	9	original	-0.256347	0.256347	0.256347
S9	10	stood	-1.091202	1.091202	1.091202
S9	11	for	-0.178729	0.178729	0.178729
S10	0	funny	1.577193	1.577193	1.577193
S10	1	heartfelt	0.803212	0.803212	0.803212
S10	2	and	0.273088	0.273088	0.273088
S10	3	endlessly	0.226613	0.226613	0.226613
S10	4	entertaining	2.835799	2.835799	2.835799
S10	5	from	0.083341	0.083341	0.083341
S10	6	beginning	0.475263	0.475263	0.475263
S10	7	to	-0.403858	0.403858	-0.403858
S10	8	end	-0.223215	0.223215	-0.223215

D Complete faithfulness results

Table 7: Complete faithfulness evaluation. Comp. is the comprehensiveness drop and suff. is the sufficiency gap.

ID	α	k	Selected words	Comp.	Suff.
S1	0.1	2	beautiful portrait	0.9782	2.1121
S1	0.2	3	beautiful portrait resilience	1.5693	1.4735
S1	0.3	4	beautiful portrait resilience moving	4.2399	0.8736
S1	0.5	6	beautiful portrait resilience moving film and	6.1418	0.3819
S2	0.1	1	waste	4.3043	1.2533
S2	0.2	2	waste money	7.4392	0.9281
S2	0.3	3	waste money this	7.2913	1.1335
S2	0.5	5	waste money this of absolute	10.4340	-0.0095
S3	0.1	2	outstanding performances	4.5409	-0.0012
S3	0.2	3	outstanding performances but	10.5901	0.1350
S3	0.3	4	outstanding performances but the	10.6039	0.0555
S3	0.5	6	outstanding performances but the are a	10.6815	-0.8054
S4	0.1	1	dull	0.2737	1.1605
S4	0.2	2	dull tedious	1.2735	0.8342
S4	0.3	3	dull tedious forgettable	9.1523	0.3967
S4	0.5	4	dull tedious forgettable completely	11.3001	0.1528
S5	0.1	2	brilliant inspired	1.3549	0.6676
S5	0.2	3	brilliant inspired and	1.8503	0.7298
S5	0.3	4	brilliant inspired and acting	5.2930	0.6618
S5	0.5	6	brilliant inspired and acting the short	10.9281	1.5858
S6	0.1	1	breathhtaking	0.3483	1.8376
S6	0.2	2	breathhtaking and	0.4023	1.9966
S6	0.3	3	breathhtaking and masterpiece	3.2961	1.5983
S6	0.5	5	breathhtaking and masterpiece storytelling a	5.8204	0.3235
S7	0.1	1	boring	0.2723	2.5498
S7	0.2	2	boring or	0.2553	1.0303
S7	0.3	3	boring or devoid	0.9360	1.5410
S7	0.5	5	boring or devoid any painfully	4.5125	1.0590
S8	0.1	2	captivating delivers	9.7611	0.3913
S8	0.2	3	captivating delivers the	10.2428	0.6670
S8	0.3	4	captivating delivers the lead	10.0259	-0.4161
S8	0.5	6	captivating delivers the lead but a	12.5431	0.7161
S9	0.1	2	hollow disappointing	1.7263	1.0374
S9	0.2	3	hollow disappointing that	2.0574	1.0839
S9	0.3	4	hollow disappointing that stood	2.6488	1.3098
S9	0.5	6	hollow disappointing that stood betrays sequel	10.9568	0.1159
S10	0.1	1	entertaining	0.2306	2.3713
S10	0.2	2	entertaining funny	1.1845	1.5115
S10	0.3	3	entertaining funny heartfelt	6.0715	0.7724
S10	0.5	5	entertaining funny heartfelt beginning and	8.8950	0.4932

E Reproducibility record

The independent implementation is stored under `bert_shap/`. Its principal files are:

- `config.json`: experiment defaults;
- `data/sentences.csv`: S1–S10 inputs and gold labels;
- `src/model_wrapper.py`: BERT inference and Equation (4);
- `src/token_aggregation.py`: Equations (11) and (12);
- `src/evaluate_faithfulness.py`: Equations (15) and (16);
- `src/make_comparison_plots.py`: sentence and global plots;
- `src/run_shap.py`: configuration, execution, and result writing;
- `tests/test_model_parity.py`: direct-model versus wrapper parity;
- `tests/test_additivity.py`: additive reconstruction and WordPiece tests.

Seven automated tests passed. The full modular run produced ten sentence rows, 104 word-level rows, forty faithfulness rows, and thirty-two plot/HTML artifacts. The 104 word-level values were compared with the pre-refactor implementation and matched exactly.

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